Lesson 1: Prime Building Blocks of Numbers

Factors, primes and composites, then two organized ways to take a number apart — the division ladder and the factor tree.

1. Factors

The **factors** of a whole number are the whole numbers dividing into it evenly: only 1, 2, 7 and 14 divide into 14, so 14 has four factors. Factors come in pairs — $14 = 1 \times 14 = 2 \times 7$ — so listing the pairs in order from 1 makes sure none are missed.

Example 1 — Listing factors, and counting them

(a) State the factors of the following.

(i) 16 (ii) 40

(b) In each case, determine the number of factors of the given whole number.

(i) 4 (ii) 12 (iii) 1 (iv) 11 (v) 28

2. Prime and Composite Numbers

A **prime number** has *exactly* two factors, 1 and itself (2, 13); a **composite number** has *more than* two factors (12, 28). The number 1 has only one factor, so it is neither prime nor composite. The number 0 is neither as well: every whole number divides into 0 (for example, $0 = 5 \times 0$), so 0 has infinitely many factors, and no product of primes can ever equal 0. To test for primality, divide by each prime up to the square root; if none divides evenly, the number is prime.

Complete the list of the first ten prime numbers:

2, 3, 5, ___, ___, ___, ___, ___, ___, ___, ...

Example 2 — Prime or composite?

Classify the following whole numbers as prime or composite.

(a) 51 (b) 59 (c) 63 (d) 91 (e) 97

3. Prime Factors

The **prime factors** of a whole number are those of its factors that are themselves prime. The factors of 18 are 1, 2, 3, 6, 9 and 18; only 2 and 3 are prime, so those are the prime factors of 18.

Example 3 — Factors, prime factors, and a product of primes

(a) State the factors of 44.

(b) State the prime factors of 44.

(c) Express 44 as a product of prime factors.

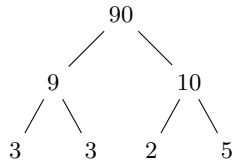
4. Prime Factorization

Prime factorization breaks a composite number down until only primes remain, and every composite number has one unique set of them. Two organized methods are the **division ladder** (divide repeatedly by the smallest prime that fits, until the quotient reaches 1; the divisors down the left are the prime factors) and the **factor tree** (split the number into any factor pair, then keep branching every composite until each branch ends in a prime). Both give the same set of primes; both are applied to 90 below.

Division Ladder

2	90
3	45
3	15
5	5
	1

Factor Tree



$$90 = 2 \times 3 \times 3 \times 5 \quad \text{or} \quad 2 \times 3^2 \times 5$$

Example 4 — The division ladder on a larger number

Use a division ladder to determine the prime factorization of 3 432.

Example 5 — The factor tree on a larger number

Use a factor tree to determine the prime factorization of 7 650.

Lesson 2: Prime Factorization at Work: GCF and LCM

Prime factorizations give a fast, reliable route to the greatest common factor and the least common multiple.

Starter. Write each of the following as a product of prime factors: 60 and 90.

1. Greatest Common Factor

The **greatest common factor** (GCF) of a group of whole numbers is the largest whole number that divides evenly into every member of the group. The GCF of 10, 15 and 25 is 5.

Example 1 — GCF by inspection

State the greatest common factor of each group.
(a) 16, 24, and 40 (b) 21 and 28 (c) 48 and 72

Inspection gets hard once a number has many factors, as in (c); prime factorization gives a shortcut. From the starter, $60 = 2 \times 2 \times 3 \times 5$ and $90 = 2 \times 3 \times 3 \times 5$. **To find the GCF, multiply together every prime factor (matching repeats) that appears in *every* factorization.**

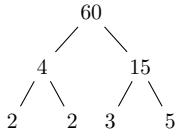
Division ladder: 60

2	60
2	30
3	15
5	5
	1

Division ladder: 90

2	90
3	45
3	15
5	5
	1

Factor tree: 60



Both numbers share one 2, one 3, and one 5, so the GCF of 60 and 90 is $2 \times 3 \times 5 = 30$. **With exponents:** the GCF uses only the primes that appear in *every* factorization, each to its *lowest* power. Here $60 = 2^2 \times 3 \times 5$ and $90 = 2 \times 3^2 \times 5$, so the GCF is $2^1 \times 3^1 \times 5^1 = 30$.

Example 2 — GCF by prime factorization

Use prime factorization to determine the greatest common factor of the given whole numbers.

(a) 126 and 210 (b) 165 and 275 (c) 180, 240, and 420 (d) 132, 220, 308, and 396

2. Least Common Multiple

The **least common multiple** (LCM) of a group of numbers is the smallest number that appears in every list of multiples. Multiples of 9 are $9, 18, 27, 36, 45, \ldots$ and of 12 are $12, 24, 36, 48, \ldots$, so the LCM of 9 and 12 is 36.

Example 3 — LCM by listing multiples

State the least common multiple of each group.

(a) 5 and 9 (b) 6, 8, and 12 (c) 15, 18, and 21

Trial and error gets slow once the numbers grow, as in (c). Prime factors make it manageable: $15 = 3 \times 5$, $18 = 2 \times 3 \times 3$, $21 = 3 \times 7$. **To build the LCM, start with all the prime factors of one number, then bring in any *extra* factors needed from the others** (factors already counted are not repeated). Here: 3 and 5 from 15, an extra 2 and 3 from 18, a 7 from 21, giving $3 \times 5 \times 2 \times 3 \times 7 = 630$. **With exponents:** the LCM uses *every* prime that appears in any of the factorizations, each to its *highest* power. Here $15 = 3 \times 5$, $18 = 2 \times 3^2$ and $21 = 3 \times 7$, so the LCM is $2 \times 3^2 \times 5 \times 7 = 630$.

Example 4 — LCM by prime factorization

Use prime factorization to determine the least common multiple of each group.

(a) 18 and 45 (b) 84 and 126 (c) 20, 45, and 75

3. GCF or LCM? Reading a Word Problem

Both tools come from the same prime factorizations, so the real work in a word problem is choosing the right one. Ask what the answer has to do.

GCF — the answer must *divide into* each number: sharing things into identical groups with nothing left over, cutting lengths into the longest equal pieces, the largest square tile that fits. The GCF is never bigger than the smallest number.

LCM — each number must *divide into* the answer: events on different cycles happening together again, buying packages until the totals match. The LCM is never smaller than the largest number.

Example 5 — GCF or LCM?

Decide whether each problem calls for the GCF or the LCM, then solve it.

(a) A florist has 48 roses and 60 tulips. She makes identical bouquets using all of the flowers. What is the greatest number of bouquets she can make, and what is in each one?

(b) Two lighthouse beacons flash together at midnight. One flashes every 12 seconds and the other every 20 seconds. How many seconds will it be until they next flash together?

Lesson 3: Perfect Squares and Perfect Cubes

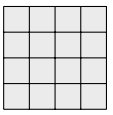
Prime factorization shows at a glance whether a number is a perfect square or a perfect cube — and hands you its root.

Starter. Write each number as a product of prime factors in exponent form: (a) 12 (b) $12^2 = 144$ (c) $12^3 = 1\,728$

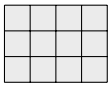
1. Prime Factorization of a Perfect Square

Numbers such as $1, 4, 9, 16, 25, 36, 49, \ldots$ — each the square of a whole number — are **perfect squares**, and $\sqrt{\ }$ means the positive, or **principal**, square root. Factor a perfect square into primes and **every prime appears an *even* number of times** — the prime factors pair up perfectly.

Concretely: a perfect square is a number of square tiles that can be arranged into one solid square. 16 tiles make a 4×4 square, so 16 is a perfect square; 12 tiles only ever make rectangles (1×12, 2×6, 3×4), so 12 is not.



$16 = 4 \times 4$ (a square)



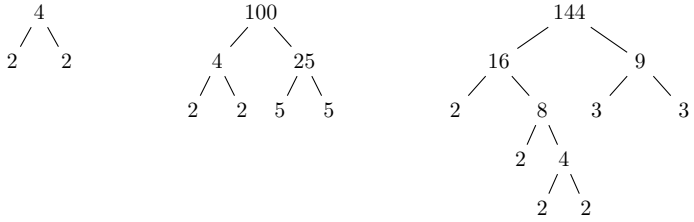
$12 = 3 \times 4$ (not a square)

Why even? A perfect square is a whole number times itself, so its prime factorization writes every prime factor down *twice*: every exponent doubles. From the starter,

$$12^2 = (2^2 \times 3) \times (2^2 \times 3) = 2^4 \times 3^2.$$

Doubling always gives an even number, so **every exponent in a perfect square is even**. Going backwards, **halve each exponent** to get the square root: $\sqrt{144} = \sqrt{2^4 \times 3^2} = 2^2 \times 3 = 12$.

Below are three perfect squares factored.



$4 = 2 \times 2$, $100 = 2 \times 2 \times 5 \times 5$, $144 = 2 \times 2 \times 2 \times 2 \times 3 \times 3$. Pairing up the factors: $\sqrt{4} = 2$, $\sqrt{100} = 2 \times 5 = 10$, $\sqrt{144} = 2 \times 2 \times 3 = 12$. If the prime factorization does *not* split into matching pairs, the number is not a perfect square.

Example 1 — A perfect square, two ways

Consider the number 15 876.

(a) Use a calculator to find the square root of 15 876.

(b) **Explain** how the prime factorization of 15 876 shows that it must be a perfect square, and use that factorization to verify your calculator answer.

Example 2 — Testing for a perfect square

In each case, use prime factorization to decide whether the number is a perfect square. If it is, state its square root.

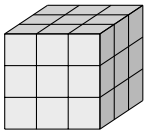
(a) 1 089

(b) 2 028

2. Prime Factorization of a Perfect Cube

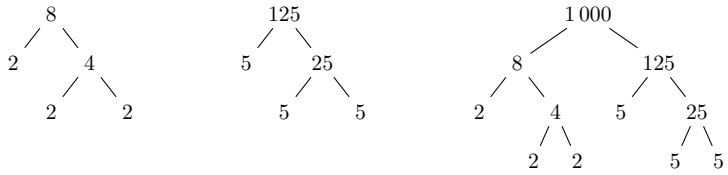
Perfect cubes of whole numbers include $1, 8, 27, 64, 125, \ldots$, with **cube roots** $1, 2, 3, 4, 5, \ldots$. In the prime factorization of a perfect cube, **every prime factor appears a number of times that is a multiple of three** — the factors group into matching triples.

Concretely: a perfect cube is a number of small cubes that can be stacked into one solid larger cube. 27 cubes make a $3 \times 3 \times 3$ cube, so 27 is a perfect cube; 18 cubes can't be stacked into a solid cube, so 18 is not.



$27 = 3 \times 3 \times 3$

Why multiples of three? Cubing writes every prime factor down *three* times, so every exponent triples: $12^3 = (2^2 \times 3) \times (2^2 \times 3) \times (2^2 \times 3) = 2^6 \times 3^3$. Going backwards, **divide each exponent by** 3 to get the cube root: $\sqrt[3]{1\,728} = \sqrt[3]{2^6 \times 3^3} = 2^2 \times 3 = 12$.



$8 = 2 \times 2 \times 2$, $125 = 5 \times 5 \times 5$, $1\,000 = 2 \times 2 \times 2 \times 5 \times 5 \times 5$. Grouping into triples: $\sqrt[3]{8} = 2$, $\sqrt[3]{125} = 5$, $\sqrt[3]{1\,000} = 2 \times 5 = 10$. If the prime factorization does not break into matching triples, the number is not a perfect cube.

Example 3 — Testing for a perfect cube

In each case, use prime factorization (a division ladder works well) to determine whether the number is a perfect cube. If it is, state its cube root.

(a) 5 832 (b) 39 304

3. Square, Cube, Both, or Neither?

Write the number in exponent form and look only at the **exponents**: all even $\Rightarrow$ a perfect square; all multiples of 3 $\Rightarrow$ a perfect cube; all even *and* multiples of 3 — that is, all multiples of 6 — $\Rightarrow$ both. For example, $64 = 2^6$ is both: $\sqrt{64} = 2^3 = 8$ and $\sqrt[3]{64} = 2^2 = 4$.

Example 4 — Square, cube, both, or neither?

Decide whether each number is a perfect square, a perfect cube, both, or neither. State each root that is a whole number.

(a) 400 (b) 216 (c) 4 096 (d) 1 200

4. Making a Perfect Square or a Perfect Cube

When an exponent is not even, the number is not a perfect square — but multiplying by the right primes fixes it. To find the **smallest** multiplier, raise each odd exponent to the *next* even number, and nothing more. (For a perfect cube, raise each exponent to the next multiple of 3.) For example, $50 = 2 \times 5^2$: the 2 has no partner, so $50 \times 2 = 100 = 2^2 \times 5^2 = 10^2$.

Example 5 — The smallest multiplier

Consider the number 72.

(a) Find the smallest whole number 72 must be multiplied by to give a perfect square, and state the square root of the result.

(b) Find the smallest whole number 72 must be multiplied by to give a perfect cube, and state the cube root of the result.

5. Squares and Cubes in Geometry

The area of a square is side × side, so the side length is $\sqrt{\text{area}}$. The volume of a cube is edge × edge × edge, so the edge length is $\sqrt[3]{\text{volume}}$. Prime factorization finds these roots exactly, and a calculator can check them.

Example 6 — Side lengths from area and volume

(a) A square poster has an area of 3 136 cm^2. Use prime factorization to find its side length.

(b) A cube-shaped block of ice has a volume of 3 375 cm^3. Use prime factorization to find the length of each edge.

Lesson 4: Rational Numbers, Irrational Numbers, and Decimal Patterns

Decimals that arrive versus decimals that wander — and how to turn a repeating decimal back into a fraction.

1. Starter: Where Does the Decimal Lead?

Convert each number to a decimal (long division or calculator) and circle the words that describe the pattern you get.

(a) $\sqrt{3}$ is (repeating / non-repeating) and (terminates / does not terminate).

(b) $\frac{17}{6}$ is (repeating / non-repeating) and (terminates / does not terminate).

(c) $\frac{17}{4}$ is (repeating / non-repeating) and (terminates / does not terminate).

2. Terminology

A **repeating decimal** has a block of digits that copies itself forever, e.g. $0.4141414\ldots = 0.\overline{41}$, with the bar over the repeating block only; a **non-repeating decimal** never falls into such a block. A **terminating decimal** has finitely many digits; a **non-terminating decimal** continues forever. Every decimal gets one label from each pair, but only three combinations ever happen: a terminating decimal is always non-repeating, so a decimal is *terminating and non-repeating*, *non-terminating and repeating*, or *non-terminating and non-repeating*.

Reading "…". When a decimal ends in "…", assume the digits continue the way they are shown: a block that repeats keeps repeating forever, and digits that follow a growing pattern (or no pattern at all) never settle into a repeating block.

Example 1 — Repeating or not, terminating or not

For each number, state whether its decimal form is repeating or non-repeating, and whether it is terminating or non-terminating.

(a) $\frac{1}{4}$

(b) $\frac{3}{11}$

(c) $\frac{5}{6}$

(d) 0.68

(e) $\sqrt{12}$

(f) $\sqrt{0.25}$

3. Rational and Irrational Numbers

A decimal that terminates, or repeats forever in a fixed block, can be rewritten as a ratio of two integers; these are the **rational numbers**. One that runs on forever *without* settling into a repeating block cannot be, and these are the **irrational numbers**. Quick test for a fraction in lowest terms: if the denominator's prime factors are only 2's and 5's the decimal terminates, otherwise it repeats — either way it is rational.

Example 2 — Rational or irrational?

Classify each number as rational or irrational and give a short reason. For each rational number, use a calculator to write it as an improper fraction in simplest form.

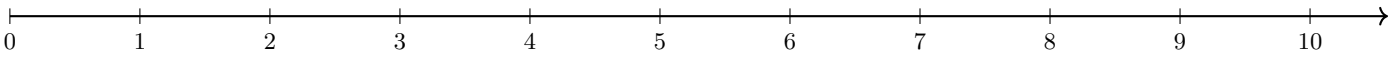
(a) 2.375 **(b)** $\sqrt{7}$ **(c)** $1.512\,512\,512\ldots$

(d) $-\sqrt{121}$ **(e)** $\sqrt{2.89}$ **(f)** $-7.12\,112\,1112\,11112\ldots$

Example 3 — Ordering irrational numbers on a number line

Place the following irrational numbers, in order, on the number line below.

(a) $\sqrt{19}$ **(b)** π **(c)** $\sqrt{90}$ **(d)** $3\sqrt{8}$ **(e)** $\sqrt{0.3}$ *($3\sqrt{8}$ means $3 \times \sqrt{8}$.)*



4. Converting a Repeating Decimal to a Fraction: Calculator Method

The TI-30XIIS has a decimal-to-fraction key, F◄►D: type in enough repeated digits to fill the screen, press ENTER, then F◄►D, and it recognizes the pattern and returns the exact fraction. It only works when the fraction's denominator is less than 1000; for any other repeating decimal, use the algebraic method in Section 5.

Example 4 — Calculator conversions

Use a calculator to convert each repeating decimal to an improper fraction in simplest form.

(a) $0.\overline{45}$ **(b)** $2.8\overline{3}$

5. Extension — Converting a Repeating Decimal to a Fraction: Algebraic Method

Step 1. Let x equal the repeating decimal. **Step 2.** Identify the repeating digits. **Step 3.** Multiply both sides of Step 1 by the power of 10 that shifts the repeating block to the **left** of the decimal point. **Step 4.** If non-repeating digits sit right after the point, multiply Step 1 instead by the power of 10 that puts the repeating block **immediately to the right** of it. **Step 5.** Subtract Step 4 from Step 3 (or Step 1 from Step 3 if there is no Step 4): the repeating tails cancel, leaving an equation to solve for x.

Example 5 — Finishing the procedure

A student is converting $3.0\overline{62}$ to a fraction. Steps 1–4 are done; complete Step 5.

Step 1. Let $x = 3.0\overline{62}$.

Step 2. The repeating digits are 62.

Step 3. To move 62 left of the decimal point, multiply both sides by $10^3 = 1000$: $1000x = 3062.\overline{62}$

Step 4. To move 62 to sit immediately right of the decimal point, multiply both sides by 10: $10x = 30.\overline{62}$

Step 5. Subtract, and solve for x. *(Complete this step.)*

Example 6 — The algebraic method from scratch

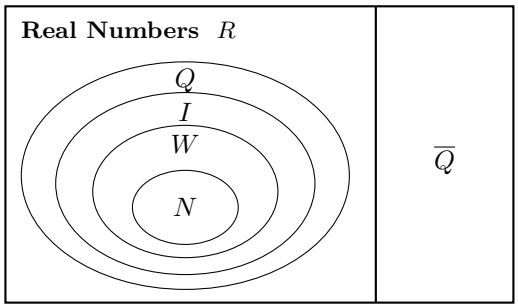
Use the algebraic procedure to convert each of the following to a fraction in simplest form.
(a) $0.\overline{45}$ **(b)** $2.8\overline{3}$

Lesson 5: Classifying the Real Number System

Sorting every real number into N, W, I, Q and $\overline{Q}$ — then using square and cube roots to decide where a number lands.

1. The Real Number System

Divide one integer by another by long division and only two things can happen: it terminates, or it settles into a block of digits that repeats forever. Either way the result is a **rational number** — a number built from a *ratio* of integers. Numbers whose decimal expansion never terminates *and* never repeats — π and $\sqrt{2}$ are the famous examples — are **irrational numbers**. Together the rationals and irrationals fill every point on the number line, forming the **real numbers**.



Natural numbers	$N = \{1, 2, 3, \dots\}$
Whole numbers	$W = \{0, 1, 2, 3, \dots\}$
Integers	$I = \{\dots, -2, -1, 0, 1, 2, \dots\}$
Rational numbers	$Q = \left\{\frac{a}{b},\ a, b \in I,\ b \neq 0\right\}$
Irrational numbers	$\overline{Q} = \{$non-terminating, non-repeating decimals$\}$
Real numbers	$R = \{Q$ and $\overline{Q}\}$

Each set above is contained inside the next: every natural number is whole, every whole number an integer, every integer rational. The irrationals sit off to the side — no irrational number is also rational.

Example 1 — Naming every set a number belongs to

(a) For each number below, list *every* set (from $N, W, I, Q, \overline{Q}, R$) that it belongs to, ordered from the largest set to the smallest set.

(i) 12 (ii) $\frac{5}{9}$ (iii) $\sqrt{11}$ (iv) -8

(b) Explain why 12 belongs to five different number sets, while -12 belongs to only three.

2. Square Roots

Every positive real number has **two** square roots, one positive and one negative. The positive root is the **principal square root**, and it is what the symbol $\sqrt{\ }$ always refers to: the square roots of 49 are 7 and -7, but $\sqrt{49} = 7$ only. The square roots of a **perfect square** are rational; otherwise they are irrational, like $\sqrt{23}$ and $-\sqrt{23}$. Mental estimates catch calculator mistakes, and rely on knowing the perfect squares.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
n^2	1	4	9	16	25	36	49	64	81	100	121	144	169	196	225

Example 2 — Estimating square roots mentally

For each of the following, (i) estimate the value mentally, to the nearest whole number; (ii) use a calculator to find the decimal approximation to the nearest tenth, and decide whether the mental estimate in (i) was reasonable.

(a) $\sqrt{53}$ (b) $3\sqrt{20}+4\sqrt{41}$ (c) $-\sqrt{7}$ (d) $\sqrt{\sqrt{150}}$

To estimate the square root of a very large (or very small) number, group its digits in *pairs* starting from the decimal point. Each pair corresponds to one digit of the square root.

Example 3 — Estimating to one significant digit

Estimate each square root to one significant digit.

(a) $\sqrt{7\,340}$ (b) $\sqrt{51\,234.6}$ (c) $\sqrt{0.005\,12}$ (d) $\sqrt{0.051\,2}$

3. Cube Roots

Every real number — positive *or* negative — has exactly **one** cube root, written $\sqrt[3]{\ }$. The cube root of a **perfect cube** is rational ($\sqrt[3]{8\,000}=20$, $\sqrt[3]{-64}=-4$); otherwise it is irrational, e.g. $\sqrt[3]{15}$.

n	1	2	3	4	5	6	7	8	9	10
n^3	1	8	27	64	125	216	343	512	729	1 000

Example 4 — Estimating cube roots mentally

For each of the following, (i) estimate the value mentally, to the nearest whole number; (ii) use a calculator to find the decimal approximation to the nearest tenth, and decide whether the mental estimate in (i) was reasonable.

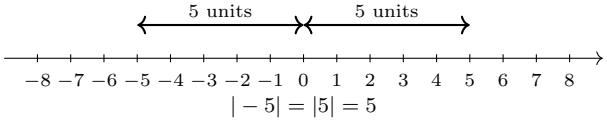
(a) $\sqrt[3]{30}$ (b) $\sqrt[3]{150}$ (c) $5\sqrt{50}-5\sqrt[3]{50}$

4. Extension — Absolute Value

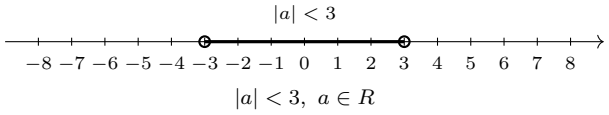
The **absolute value** of a real number is the principal square root of its square:

$$\text{absolute value of } 8=\sqrt{(8)^2}=\sqrt{64}=8, \qquad \text{absolute value of } -8=\sqrt{(-8)^2}=\sqrt{64}=8.$$

For a real number a we write the absolute value as $|a|$, so $|8|=8$ and $|-8|=8$; geometrically $|a|$ is the *distance* of a from 0. To graph $|a|<3,\ a\in R$ we need every point less than 3 units from 0, so the endpoints are open.



$|-5|=|5|=5$



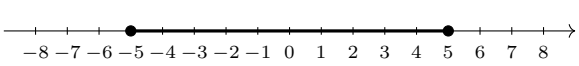
$|a|<3,\ a\in R$

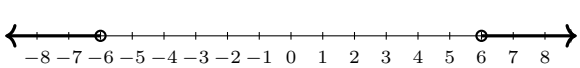
Example 5 — Absolute value and its graphs

Evaluate each of the following.

(a) $|-9|+|9|$ (b) $|2-9|$ (c) $-|-\sqrt{100}|$

State the absolute value inequality represented by each graph.

(d) 

(e) 

Lesson 6: Roots of Real Numbers

Square roots, cube roots and roots of any index — radical vocabulary, and the product and quotient rules.

1. Recap from Lesson 5

$\sqrt{\ }$ always means the **principal** (positive) square root: 36 has two square roots, ± 6, but $\sqrt{36} = 6$ only. Every real number has exactly one cube root, so negatives are fine there: $\sqrt[3]{-8} = -2$ since $(-2)^3 = -8$.

2. Roots of Other Degrees

Square roots and cube roots extend naturally to a root of any whole-number degree. Complete each statement below.

$3 \times 3 = 9$, so a square root of 9 is _____, or $\sqrt{9} =$ _____

$4 \times 4 \times 4 = 64$, so the cube root of 64 is _____, or $\sqrt[3]{64} =$ _____

$3 \times 3 \times 3 \times 3 = 81$, so a fourth root of 81 is _____, or $\sqrt[4]{81} =$ _____

$2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 128$, so _____

Example 1 — Evaluating radicals mentally

Mentally evaluate, where possible, using the real number system.

(a) $\sqrt{121}$

(b) $\sqrt[3]{-125}$

(c) $\sqrt[4]{6\,561}$

(d) $\sqrt[5]{\dfrac{1}{243}}$

(e) $\sqrt[4]{-81}$

(f) $3\sqrt[3]{8}$

3. Evaluating Radicals with a Calculator

The TI-30XIIS has an $\sqrt[x]{\ }$ (xth root) function, [2nd] [^], that evaluates a radical of any index directly. For $\sqrt[5]{161\,051}$: enter the index [5]; press [2nd] [^] (the xth-root key $\sqrt[x]{\ }$); enter the radicand [161051]; press [ENTER]. The display shows 11 — the index is keyed *before* the root symbol.

Example 2 — Using the xth-root key

(a) Use a calculator to evaluate each of the following exactly.

(i) $\sqrt[5]{243}$

(ii) $\sqrt[7]{-128}$

(iii) $-2\sqrt[4]{10\,000}$

(iv) $\sqrt[3]{\dfrac{64}{27}}$

(b) Use a calculator to evaluate each of the following to the nearest hundredth.

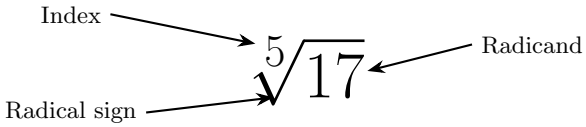
(i) $\sqrt[5]{200}$

(ii) $\sqrt[6]{0.7}$

(iii) $\dfrac{3}{4}\sqrt[4]{500}$

4. Radical Vocabulary

Any expression of the form $\sqrt[n]{x}$, where $n \in N$ and $n \geq 2$, is a **radical**: $\sqrt{19}$, $\sqrt[3]{54}$, $\sqrt[5]{7}$, $\sqrt[6]{2\,000\,000}$. The symbol $\sqrt{\ }$ is the **radical sign**, x is the **radicand**, and n is the **index**, which tells you how many times the root is used as a *factor* to produce x: $\sqrt[5]{32} = 2$ because $2 \times 2 \times 2 \times 2 \times 2 = 32$ (five factors of 2). **If the index is even, the radicand must be non-negative**, or the radical is not a real number; a missing index means 2.



Example 3 — Naming the index and the radicand

Identify the index and the radicand in each of the following.

(a) $\sqrt[7]{200}$ (b) $\sqrt{85}$ (c) $4\sqrt[5]{-\frac{1}{9}}$

5. Combining Radicals: Products and Quotients

Using 16 and 9:

$\sqrt{16} \times \sqrt{9} = \sqrt{16 \times 9}$ $\quad$ $\sqrt{16} + \sqrt{9}$ is **not** equal to $\sqrt{16+9}$

$\sqrt{16} \div \sqrt{9} = \sqrt{16 \div 9}$ $\quad$ $\sqrt{16} - \sqrt{9}$ is **not** equal to $\sqrt{16-9}$

Two general rules follow. **(i)** The product (or quotient) of the roots of two numbers equals the root of their product (or quotient): $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$ where $a, b \geq 0$, and $\dfrac{\sqrt{a}}{\sqrt{b}} = \sqrt{\dfrac{a}{b}}$ where $a \geq 0,\ b > 0$. **(ii)** The sum (or difference) of the roots is **NOT** equal to the root of the sum (or difference).

Example 4 — True or false

State whether each statement is true or false.

(a) $\sqrt{4} \times \sqrt{11} = \sqrt{44}$ (b) $\dfrac{\sqrt{63}}{\sqrt{7}} = \sqrt{8}$ (c) $\sqrt{25+16} = \sqrt{25} + \sqrt{16}$

Example 5 — Writing a product or quotient as a single radical

Write the following as a single radical in the form $\sqrt{x}$.

(a) $\sqrt{12} \times \sqrt{3}$ (b) $\sqrt{11} \times \sqrt{5}$ (c) $\dfrac{\sqrt{72}}{\sqrt{8}}$ (d) $\dfrac{\sqrt{\sqrt{256}}}{\sqrt{2}}$

Example 6 — Splitting a radical into a product

Express as a product of two radicals.

(a) $\sqrt{85}$ (b) $\sqrt{98}$ (c) $\sqrt{32}$

Lesson 7: Entire and Mixed Radicals

Run the product rule backwards — pull the largest perfect square out in front of the root, and keep Pythagorean answers exact.

1. Splitting a Radical Apart

Since $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$ (for $a, b \geq 0$) and $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$ (for $a \geq 0$, $b > 0$), *any* square root can be rewritten as a product or quotient of two smaller square roots, once we choose a sensible split.

Example 1 — splitting one root into two

Rewrite each as a product or quotient of two square roots.

(a) $\sqrt{63} = \sqrt{9 \times __}$
$= \sqrt{__} \times \sqrt{__}$

(b) $\sqrt{32} = \sqrt{16 \times __}$
$= \sqrt{__} \times \sqrt{__}$

(c) $\sqrt{\frac{17}{9}} = \frac{\sqrt{__}}{\sqrt{__}}$

2. Entire Radicals and Mixed Radicals

Use a calculator to approximate each to five decimal places: (i) $\sqrt{180} =$ _____ (ii) $2\sqrt{45} =$ _____ (iii) $6\sqrt{5} =$ _____. What do you notice? _____

Fill in the blanks to see algebraically why all three expressions represent the same number.

$$\sqrt{180} = \sqrt{4 \times 45} = \sqrt{__} \times \sqrt{__} = _____ \qquad \sqrt{180} = \sqrt{36 \times 5} = \sqrt{__} \times \sqrt{__} = _____$$

A radical of the form $\sqrt[n]{b}$ is an **entire (pure) radical** — the number sits alone under the root sign: $\sqrt{31}$, $\sqrt{150}$, $\sqrt[3]{19}$. One of the form $a\sqrt[n]{b}$ is a **mixed radical** — a coefficient sits outside: $\frac{5}{6}\sqrt{3}$, $7\sqrt{11}$, $-4\sqrt[3]{19}$. Every mixed radical with a positive coefficient can be rewritten as an entire radical (a negative coefficient stays outside the root: $-7\sqrt{5} = -\sqrt{245}$); an entire square root can be rewritten as a mixed radical exactly when its radicand has a perfect-square factor.

3. Converting an Entire Radical to a Mixed Radical

Identify the *largest* perfect square dividing the radicand, then use the product rule to split it off as a whole number outside the root. Complete the steps to convert $\sqrt{200}$:

$$\sqrt{200} = \sqrt{__ \times 2} = \sqrt{__} \times \sqrt{2} = _____$$

1^2	2^2	3^2	4^2	5^2	6^2	7^2	8^2	9^2	10^2	11^2	12^2	13^2	14^2	15^2
1	4	9	16	25	36	49	64	81	100	121	144	169	196	225

Reminder: a mixed radical is complete only once no perfect-square factor remains underneath; pull out a smaller square and you must repeat the process on what is left.

Example 2 — entire → mixed

Convert each radical to a mixed radical in simplest form.

(a) $\sqrt{72}$ (b) $\sqrt{63}$ (c) $\sqrt{288}$

Example 3 — with a coefficient, and with a fraction underneath

Convert each radical to a mixed radical in simplest form.

(a) $5\sqrt{128}$ (b) $\frac{2}{3}\sqrt{252}$ (c) $\sqrt{\frac{11}{16}}$

4. Converting a Mixed Radical to an Entire Radical

To reverse the process, square the outside coefficient to bring it back under the root sign, then multiply by the existing radicand. Complete the steps to convert $4\sqrt{11}$:

$$4\sqrt{11} = \sqrt{__} \times \sqrt{11} = \sqrt{__ \times 11} = _____$$

Example 4 — mixed → entire

Convert each mixed radical to an entire radical.
(a) $3\sqrt{8}$ (b) $5\sqrt{6}$ (c) $9\sqrt{5}$ (d) $\frac{5}{2}\sqrt{8}$ (e) $0.6\sqrt{75}$ (f) $-7\sqrt{5}$

5. Comparing Radicals

To compare mixed radicals without a calculator, convert them all to entire radical form and compare the numbers under the root signs directly.

Example 5 — ordering without a calculator

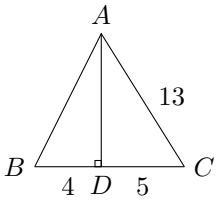
Arrange in order from greatest to least: i) $6\sqrt{3}$ ii) $4\sqrt{5}$ iii) $\sqrt{40}$ iv) $3\sqrt{6}$ v) $2\sqrt{11}$.

6. Radicals in the Pythagorean Theorem

Lengths found with the Pythagorean Theorem are frequently irrational, so answers are commonly left in exact mixed radical form rather than rounded to a decimal.

Example 6 — exact side length

In the diagram, AD is the altitude from A to BC. Use the Pythagorean Theorem to find the exact length of AB.



(a) Express AB as an exact value in simplest mixed radical form.

(b) Express AB as a decimal, rounded to the nearest hundredth.

Lesson 8: Entire and Mixed Radicals: Index 3 and Higher

The same idea for cube roots and beyond — pull out the largest perfect nth power. (Extension: radicals with variables.)

In Lesson 7, $\sqrt{200} = \sqrt{100 \times 2} = 10\sqrt{2}$: the largest perfect *square* factor came out in front of the root. For a cube root, pull out the largest perfect *cube* factor; for a fourth root, the largest perfect fourth power; and so on.

1. Entire to Mixed Radicals (Index 3 and Higher)

An entire radical with index n becomes a mixed radical once the largest perfect nth-power factor is pulled out of the radicand — for index 3, the largest **perfect cube** dividing it.

n	2	3	4	5	6	7	8	9	10
n^2	4	9	16	25	36	49	64	81	100
n^3	8	27	64	125	216	343	512	729	1 000
n^4	16	81	256	625	1 296	2 401	4 096	6 561	10 000
n^5	32	243	1 024	3 125	7 776	16 807	32 768	59 049	100 000

Worked demonstration. $\sqrt[3]{250} = \sqrt[3]{125 \times 2} = \sqrt[3]{125} \times \sqrt[3]{2} = 5\sqrt[3]{2}$, choosing $125 = 5^3$ as the *largest* perfect cube dividing 250. Any index works the same way.

Example 1 — higher-index entire → mixed

Convert to mixed radicals in simplest form.
(a) $\sqrt[3]{1080}$ (b) $\sqrt[5]{3072}$ (c) $\sqrt[3]{-135}$

2. Mixed to Entire Radicals (Index 3 and Higher)

To reverse the process, convert the coefficient into a radical of the *same index* by raising it to the power n, then combine the two radicals under one root. For example, $\frac{1}{3}\sqrt[3]{54} = \sqrt[3]{\left(\frac{1}{3}\right)^3} \times \sqrt[3]{54} = \sqrt[3]{\frac{1}{27}} \times \sqrt[3]{54} = \sqrt[3]{\frac{54}{27}} = \sqrt[3]{2}$.

Watch the sign. With an *odd* index a negative sign travels freely in or out of the radical, since an odd power of a negative number is still negative. With an *even* index it cannot: an even root of a negative number is not real, so a negative coefficient must stay outside.

Example 2 — higher-index mixed → entire

Convert to entire radicals.

(a) $3\sqrt[4]{2}$ (b) $-3\sqrt[3]{11}$ (c) $\frac{3}{4}\sqrt[3]{32}$ (d) $-2\sqrt[5]{5}$

3. Extension: Radicals With Variables

The same two conversions work when the radicand contains variables — track exponents instead of values. *Assume every variable is non-negative, so no absolute-value signs are needed.* Dividing the exponent by the index gives the exponent that comes out:

$$\sqrt{x^{10}} = x^5, \qquad \sqrt{y^{14}} = y^7, \qquad \sqrt{a^{12}b^4} = a^6b^2, \qquad \sqrt[3]{x^{18}} = x^6, \qquad \sqrt[3]{y^9} = y^3$$

When the exponent does *not* divide evenly by the index, split it into the largest multiple of the index plus a remainder: $\sqrt{x^7} = \sqrt{x^6 \times x} = \sqrt{x^6} \times \sqrt{x} = x^3\sqrt{x}$.

Example 3 — variable entire → mixed

Convert the following entire radicals to mixed radicals in simplest form.

(a) $\sqrt{a^9}$ (b) $\sqrt{t^{11}}$ (c) $\sqrt[3]{x^8}$ (d) $\sqrt[3]{x^{10}}$

(e) $\sqrt{x^8y^7}$ (f) $\sqrt{50x^5}$ (g) $\sqrt{48y^5z^6}$ (h) $\sqrt[3]{54x^7y^{10}}$

Example 4 — variable mixed → entire

Convert the following mixed radicals to entire radicals.

(a) $3\sqrt{x^5}$ (b) $b^3\sqrt{b}$ (c) $x^4\sqrt{xy}$ (d) $2pq^2\sqrt[3]{3r^5}$

Lesson 9: Number Unit Review

One pass through the whole unit: prime factorization, GCF and LCM, perfect powers, the real number system, and radicals.

1. Prime Factorization, GCF, and LCM

Every whole number greater than 1 breaks apart uniquely into a product of primes — its **prime factorization**. From two such factorizations, the **greatest common factor** (GCF) is the product of the primes they *share*, each taken to the *lowest* power present, and the **least common multiple** (LCM) is the product of every prime in *either* one, each taken to the highest power present.

Example 1 — Reading prime factors off a large number

Find each prime factorization first, then answer the question.

(a) Which of the following numbers is *not* a prime factor of 2 574?
A. 3 B. 11 C. 13 D. 19

(b) How many numbers in the list 5, 13, 17, 15 are prime factors of 4 290?
A. 1 B. 2 C. 3 D. 4

(c) The sum of all the prime factors of 43 010 is _____.

Example 2 — The LCM written as a product of primes

The least common multiple of 10 and 21 can be written as a product of prime factors in the form $a \times b \times c \times d$, where a, b, c, d are primes with $a < b < c < d$. The value of c is A. 2 B. 3 C. 5 D. 210

Example 3 — The GCF of two four-digit numbers

The greatest common factor of 3 003 and 3 135 is _____.

Example 4 — Reasoning backwards from a GCF

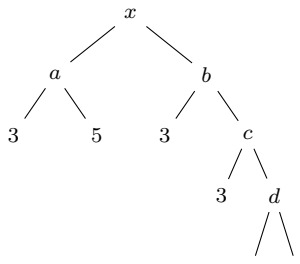
The least common multiple of 9 and 20 is equal to the greatest common factor of the numbers P and Q. Which of the following statements must be false?
A. P is a multiple of 5. B. Q is a multiple of 45. C. P could be less than 150. D. Q could be greater than 2 000.

2. Perfect Squares, Perfect Cubes, and the Factor Tree

A number is a **perfect square** when every prime in its factorization appears to an *even* power, and a **perfect cube** when every exponent is a *multiple of three*. A partly filled factor tree still works: what is missing is forced by the perfect-power condition imposed.

Example 5 — A partly completed factor tree

The tree below shows the partial prime factorization of a number x. Note that the unknown d branches into exactly two primes.



(a) If x is a perfect square, then the minimum value of d is
A. 5 B. 9 C. 15 D. 25

(b) If x is a perfect cube, then the minimum value of x is _____.

3. The Real Number System

Every real number belongs to at least one of N (natural), W (whole), I (integer), Q (rational) or $\overline{Q}$ (irrational), and which ones usually comes down to the decimal expansion: terminating or repeating means rational, everything else irrational.

Example 6 — What an irrational decimal looks like

A certain number is irrational. Its decimal representation is
A. terminating and repeating B. terminating and non-repeating C. non-terminating and repeating D. non-terminating and non-repeating

Example 7 — Sorting four numbers into rational or irrational

Which of the following are rational numbers?

$$\text{I} \quad 2.02002000200002\ldots \qquad \text{II} \quad \sqrt[3]{\frac{27}{125}} \qquad \text{III} \quad \sqrt{0.09} \qquad \text{IV} \quad 0.\overline{47}$$

A. III and IV only B. II, III, and IV only C. I, II, III, and IV D. some other combination of I, II, III, and IV

Example 8 — A repeating decimal back to a fraction

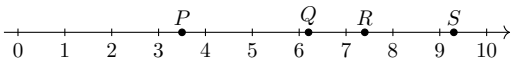
The rational number $2.\overline{36}$ can be written as an improper fraction in simplest form $\frac{c}{d}$. The value of c is A. 26 B. 13 C. 11 D. 9

4. Ordering and Estimating Radicals

A radical can be pinned between consecutive perfect powers, which is usually enough to place it on a number line. To compare radicals with *different* indices, evaluate exactly where you can and estimate the rest between perfect powers.

Example 9 — Estimating a sum of two square roots

M and N are fixed irrational numbers satisfying $20 < M < 30$ and $5 < N < 6$. The value of $\sqrt{M} + \sqrt{N}$ is best represented on the number line by A. P B. Q C. R D. S



Example 10 — Comparing radicals with different indices

Consider the following numbers: $\sqrt[3]{64}$, $\sqrt[4]{81}$, $\sqrt{17}$, $\sqrt[5]{243}$. The largest of these numbers is A. $\sqrt{17}$ B. $\sqrt[4]{81}$ C. $\sqrt[3]{64}$ D. $\sqrt[5]{243}$